

NASA AI4LS • SUMMER PROGRESS PRESENTATION

# Mixture-of-Experts Models for *Spaceflight Transcriptomic Classification*

Can organ specialists help an RNA model learn patterns useful for biological decisions?

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Computer Science + Applied Mathematics

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Berkeley AI Research • Automated  
patient-speech processing pipeline

## PROJECT EVOLUTION

A positive two-expert test led to a more ambitious *eight-expert model*

### PREVIOUS SEMESTER

Human vs. mouse

## 2 experts

Species specialists improved gene reconstruction.



### THIS SUMMER

Human organ specialists

|        |       |       |         |
|--------|-------|-------|---------|
| Brain  | Heart | Liver | Lung    |
| Muscle | Skin  | Colon | Adipose |

### HOSPITAL IDEA

One shared model, eight specialists, and a “front desk” that chooses an expert automatically.

### TRAINING

Hide gene values → fill them in.

### LATER TEST

Spaceflight response → or normal/control?

## KEY RESULT

# Organ specialists make *fewer errors* reconstructing hidden genes

## Training · fill in hidden gene values



**821**

new samples

**63**

separate studies

**≈97%**

benefit retained  
automatically

## Automatic routing keeps the benefit

**≈97%**

of the improvement remained when the model chose the organ expert itself.

- No organ label is needed when a new RNA sample arrives.
- The result held across three training runs and eight organs.

**Bottom line:** specialization helped, and the model could choose the specialist automatically.

Evidence boundary: the 821-sample result is externally evaluated and positive in all three training runs, but still requires one completely untouched confirmation cohort.

## INDEPENDENT DIAGNOSIS · MAIN LIMITATION

# Reconstruction allowed an *average-value shortcut*

### TEXT PREDICTION

#### A missing word needs context

The surrounding sentence helps reveal the meaning.

“The astronaut floated in **orbit**.”

The model must use nearby words.

≠

### RNA RECONSTRUCTION

#### A missing value has an easy shortcut

Predicting a gene’s typical value can already reduce error.

“Gene value = **usual average**”

The model may miss small changes linked to biological response.

### WHAT WE FOUND

The model could lower reconstruction error by learning typical values. That was not enough for response classification, so the specialist gain did not transfer.

## IMPACT

The summer produced both a positive result and a *tested boundary*

### BEFORE SUMMER

A broad hypothesis: biological specialists may improve RNA models.



### AFTER SUMMER

#### **Tested on new studies**

Organ specialists improve reconstruction across separate datasets.

#### **Automatically usable**

Routing preserves nearly all of the specialist benefit.

#### **Clear limit**

Better reconstruction does not guarantee better response classification.

We now know where specialization *earns its complexity*, where its current benefit stops, and what to test next.

## FUTURE DIRECTIONS

Align the training task—then test it where there is *room to improve*

### 01 · CONFIRM

#### Repeat the strongest result

Use one completely untouched dataset to confirm the reconstruction benefit.

### 02 · ALIGN

#### Train closer to the final question

Emphasize subtle biological differences within the same tissue—not only typical gene and tissue patterns.

### 03 · TEST

#### Use a stronger biological test

Build a larger labeled dataset across several organs, with room to detect improvement.

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#### ADVANCE ONLY IF

The model beats raw measurements and standard methods.

Organ specialists beat an equally capable shared model.

The next step is to align what the model learns with downstream tasks that can deliver practical biological insight.